Dynamic Hedging and Sequential Risk-Management Engine

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1 Context

At the time of writing, I have an active 11-leg sports wager placed at 672.927x odds (initial stake: Php 44.44; max payout: Php 29904.88) to enjoy more watching the FIFA World Cup 2026 Group Stage. Before the last day of the group stage (June 28, 2026), 8 out of the 11 legs resolved successfully, leaving 3 more events yet to be played out. The remaining legs are Austria to finish 2nd in group J, Portugal and Colombia to finish 1st and 2nd respectively in group K, and England to win group L.

With that, the three hedges I can do in sequence are first, England to not win group L, second is Portugal to not win their last match against Colombia (this event oppose the original leg), and lastly, Austria to not finish 2nd in group J.

2 The Opportunity

Fortunately, the 3 remaining legs are sequential and hedging the three legs is easier than when all games are simultaneous and capital do not need to be locked up all at one. Hedging is a risk-management strategy where wagers are placed on an opposing outcome from the original to either guarantee profit or minimize potential losses regardless of the matches' outcomes. The sequence of group matches are group L to play first, group K is next, and group J will play last. Considering all scenarios, there are 4 outcomes that I can end up with: full wager win or any of the three legs' hedges win.

3 The Engine

Rather than guessing arbitrary hedge amounts, I programmed a decision model in Python to easily solve for step-by-step wager sizing across varying risk appetites:

- Strategy A (Absolute Guaranteed Equal Profit): This forces the payout at every single outcome to be completely identical. Whether the full wager wins or any of the hedges win, the net profit is just the same.
- Strategy B (Confidence-Weighted Manual Sizing): Instead of an automated calculation of profit across all games, I can manually assign individual custom payout targets to each of the matches based on my confidence of them hitting.
- Strategy C (Target-Bound Split Optimization): I can input a minimum profit I want for the full wager and a minimum profit for the hedge.

- Strategy D (Budget-Capped Hedging): I can decide how much I want to walk away with if the full wager wins. The max hedge budget for the three legs is maximum payout from the full wager minus my decided amount.

Whenever necessary, the engine will also tell me if my set amounts are mathematically feasible or not. If feasible, the engine will return the exact hedge sizing for the sequence of events, otherwise, it will return why it is not.

4 The Code and Execution

```
# INITIAL SETUP
initial_stake = 44.44
full_payout = 29904.88

# EVENT ODDS (AT THE TIME OF WRITING)
england_not_to_win_odds = 6
colombia_to_win_group = 1.98
austria_not_second = 3.915

# SEQUENTIAL HEDGE ODDS
odds_1 = england_not_to_win_odds   # Event 1: England not to win group L
odds_2 = colombia_to_win_group  # Event 2: Colombia to win/draw (Colombia finish 1st and Portugal finish 2nd in group K)
odds_3 = austria_not_second    # Event 3: Austria not to finish 2nd in group J
```

Listing 1: Initial Setup

4.1 Strategy A: Absolute Guaranteed Equal Profit

```
# BACKWARD INDUCTION FORMULA

# Event 3 (Austria): We want the hedge payout to equal the full payout
hedge_3 = full_payout / odds_3
# Value of the slip heading into Event 3
value_entering_3 = full_payout - hedge_3

# Event 2 (Portugal): We want the hedge payout to equal the value of entering Event 3
hedge_2 = value_entering_3 / odds_2
# Value of the slip heading into Event 2
value_entering_2 = value_entering_3 - hedge_2

# Event 1 (England): We want the hedge payout to equal the value of entering Event 2
hedge_1 = value_entering_2 / odds_1
# Value of the slip right now
value_entering_1 = value_entering_2 - hedge_1

guaranteed_profit = value_entering_1 - initial_stake

print("SEQUENTIAL HEDGING STRATEGY")
print(f"Guaranteed Net Profit: Php {guaranteed_profit:,.2f}\n")

print("ACTION PLAN")

# Event 1 Output
print("STEP 1: MATCH 1 (England)")
```

```python
print(f"Action: Place a hedge of Php {hedge_1:,.2f} at {odds_1}x odds.")
profit_1 = (hedge_1 * odds_1) - hedge_1 - initial_stake
print(f"If it hits (England fails): You collect Php {hedge_1 * odds_1:,.2f} & STOP. Profit: Php {profit_1:,.2f}")
print(f"If it loses (England wins): Slip survives. Proceed to Step 2.\n")

# Event 2 Output
print("STEP 2: MATCH 2 (Portugal)")
print(f"Action: Place a hedge of Php {hedge_2:,.2f} at {odds_2}x odds.")
profit_2 = (hedge_2 * odds_2) - hedge_1 - hedge_2 - initial_stake
print(f"If it hits (Portugal fails): You collect Php {hedge_2 * odds_2:,.2f} & STOP. Profit: Php {profit_2:,.2f}")
print(f"If it loses (Portugal wins): Slip survives. Proceed to Step 3.\n")

# Event 3 Output
print("STEP 3: MATCH 3 (Austria)")
print(f"Action: Place a hedge of Php {hedge_3:,.2f} at {odds_3}x odds.")
profit_3 = (hedge_3 * odds_3) - hedge_1 - hedge_2 - hedge_3 - initial_stake
profit_slip = full_payout - hedge_1 - hedge_2 - hedge_3 - initial_stake
print(f"If it hits (Austria fails): You collect Php {hedge_3 * odds_3:,.2f}. Profit: Php {profit_3:,.2f}")
print(f"If it loses (Austria is 2nd): PARLAY WINS Php {full_payout:,.2f}! Profit: Php {profit_slip:,.2f}")
```

Listing 2: Strategy A: Absolute Guaranteed Equal Profit

Here is the tabulated output of Strategy A:

Step	**Target**	**Hedge Amount**	**Odds**	**Conditional Trigger Condition**	**Net Profit**
1	Match 1 (England)	Php 1,836.79	6.000x	England fails to win group	**Php 9,139.49**
2	Match 2 (Portugal)	Php 11,245.63	1.980x	England wins; Portugal fails to win match	**Php 9,139.49**
3	Match 3 (Austria)	Php 7,638.54	3.915x	Portugal wins; Austria fails to secure 2nd	**Php 9,139.49**
4	Full	—	—	Austria secures 2nd (Full wager successfully clears)	**Php 9,139.49**

Table 1: Absolute Guaranteed Equal Profit Setup

Action Plan Summary (Strategy A):

- **Step 1 (England):** Place an initial hedge of **Php 1,836.79** at $6x$ odds. If it hits, collect Php 11,020.71 and *STOP*. If it loses, the slip survives; proceed to Step 2.
- **Step 2 (Portugal):** Place a conditional hedge of **Php 11,245.63** at $1.98x$ odds. If it hits, collect Php 22,266.34 and *STOP*. If it loses, the slip survives; proceed to Step 3.
- **Step 3 (Austria):** Place a final conditional hedge of **Php 7,638.54** at $3.915x$ odds. If it hits, collect Php 29,904.88. If it loses, Austria secures second place and the primary parlay clears the maximum payout of **Php 29,904.88**.
- The total net profit at any step is **Php 9,139.49**, when all prior hedges are subtracted.

4.2 Strategy B: Confidence-Weighted Manual Sizing

```python
# CONFIDENCE-WEIGHTED TARGETS
# Set independent safety nets based on confidence!
target_profit_1 = 2500.00  # High confidence in England? Hedge lighter.
target_profit_2 = 6000.00  # Lower confidence? Hedge heavier.
target_profit_3 = 5000.00  # Lower confidence? Hedge heavier.

# WEIGHTED HEDGING FORMULA
# The math isolates each match based on target and prior sunk costs.

# Match 1
hedge_1 = (target_profit_1 + initial_stake) / (odds_1 - 1.0)

# Match 2
hedge_2 = (target_profit_2 + initial_stake + hedge_1) / (odds_2 - 1.0)

# Match 3
hedge_3 = (target_profit_3 + initial_stake + hedge_1 + hedge_2) / (odds_3 - 1.0)

total_hedge_budget = hedge_1 + hedge_2 + hedge_3
profit_slip = full_payout - total_hedge_budget - initial_stake

print("CONFIDENCE-WEIGHTED HEDGING")
print(f"If Slip WINS, your Profit: Php {profit_slip:,.2f}\n")

print("ACTION PLAN")
print(f"MATCH 1 (England - Safety Net: Php {target_profit_1:,.0f})")
print(f"Action: Hedge Php {hedge_1:,.2f} @ {odds_1}x")

print(f"\nMATCH 2 (Portugal - Safety Net: Php {target_profit_2:,.0f})")
print(f"Action: Hedge Php {hedge_2:,.2f} @ {odds_2}x (If Match 1 on slip survives)")

print(f"\nMATCH 3 (Austria - Safety Net: Php {target_profit_3:,.0f})")
print(f"Action: Hedge Php {hedge_3:,.2f} @ {odds_3}x (If Match 2 on slip survives)")
```

Listing 3: Confidence-Weighted Manual Sizing

Here is the tabulated output of Strategy B:

Step	Target	Hedge Amount	Odds	Conditional Trigger Condition	Safety Net Net Profit
1	Match 1 (England)	Php 508.89	6.000x	England fails to win group stage	**Php 2,500.00**
2	Match 2 (Portugal)	Php 6,687.07	1.980x	England wins; Portugal fails to win match	**Php 6,000.00**
3	Match 3 (Austria)	Php 4,199.11	3.915x	Portugal wins; Austria fails to secure 2nd	**Php 5,000.00**
4	Full	—	—	Austria secures 2nd (Parlay successfully clears)	**Php 18,465.38**

Table 2: Confidence-Weighted Manual Sizing

Action Plan Summary (Strategy B):

- **Match 1 (England - Safety Net: Php 2,500):** Place an initial hedge of **Php 508.89** at $6x$ odds. If it hits, you collect your targeted safety net and *STOP*. If it loses, the slip survives; proceed to Match 2.
- **Match 2 (Portugal - Safety Net: Php 6,000):** Place a conditional hedge of **Php 6,687.07** at $1.98x$ odds. If it hits, you collect your targeted safety net and *STOP*. If it loses, the slip survives; proceed to Match 3.
- **Match 3 (Austria - Safety Net: Php 5,000):** Place a final conditional hedge of **Php 4,199.11** at $3.915x$ odds. If it hits, you collect your targeted safety net. If it loses, Austria secures second place and the primary slip clears the maximum optimized profit of **Php 18,465.38**.

4.3 Strategy C: Target-Bound Split Optimization

```
# Flexible minimum targets
min_fullslip_profit = 18000.00
min_hedge_profit  = 5100.00

# FEASIBILITY CALCULATIONS

# 1. Calculate max slip profit if we guarantee `min_hedge_profit`
hedge_1 = (min_hedge_profit + initial_stake) / (odds_1 - 1.0)
hedge_2 = (min_hedge_profit + initial_stake + hedge_1) / (odds_2 - 1.0)
hedge_3 = (min_hedge_profit + initial_stake + hedge_1 + hedge_2) / (odds_3 - 1.0)

total_hedge_budget = hedge_1 + hedge_2 + hedge_3
max_possible_fullslip_profit = full_payout - total_hedge_budget - initial_stake

# 2. Calculate max hedge profit if we guarantee `min_fullslip_profit`
ratio_1 = 1.0
ratio_2 = odds_1 / (odds_2 - 1.0)
ratio_3 = ratio_2 * (odds_2 / (odds_3 - 1.0))
total_ratio = ratio_1 + ratio_2 + ratio_3

max_budget_allowed = full_payout - min_fullslip_profit - initial_stake
max_possible_hedge_profit = (max_budget_allowed * (odds_1 - 1.0) / total_ratio) - initial_stake


# OUTPUT
print("FEASIBILITY CHECKER")
if max_possible_fullslip_profit >= min_fullslip_profit:
    print("MATHEMATICALLY FEASIBLE\n")

    print("SCENARIO A: Maximize Slip Upside")
    print(f"If you secure the exact minimum hedge profit of Php {min_hedge_profit:,.2f},")
    print(f"your slip profit will be Php {max_possible_fullslip_profit:,.2f}.")
    print(f"(This exceeds your slip target by Php {max_possible_fullslip_profit - min_fullslip_profit:,.2f})")
    print(f"Total Hedge Cost required: Php {total_hedge_budget:,.2f}")

    print(f"MATCH 1: Bet Php {hedge_1:,.2f} @ {odds_1}x")
    print(f"MATCH 2: Bet Php {hedge_2:,.2f} @ {odds_2}x (If Match 1 slip survives)
```

```
")
    print(f"MATCH 3: Bet Php {hedge_3:,.2f} @ {odds_3}x (If Match 2 slip survives)\n")


    # Calculate action plan stakes for Scenario B
    h1_b = (max_possible_hedge_profit + initial_stake) / (odds_1 - 1.0)
    h2_b = (max_possible_hedge_profit + initial_stake + h1_b) / (odds_2 - 1.0)
    h3_b = (max_possible_hedge_profit + initial_stake + h1_b + h2_b) / (odds_3 - 1.0)

    print("SCENARIO B: Maximize Hedge Safety Net")
    print(f"If you secure the exact minimum slip profit of Php {min_fullslip_profit:,.2f},")
    print(f"your hedge profit will be Php {max_possible_hedge_profit:,.2f}.")
    print(f"(This exceeds your hedge target by Php {max_possible_hedge_profit - min_hedge_profit:,.2f})")
    print(f"Total Hedge Cost required: Php {max_budget_allowed:,.2f}")

    print(f" > MATCH 1: Bet Php {h1_b:,.2f} @ {odds_1}x")
    print(f" > MATCH 2: Bet Php {h2_b:,.2f} @ {odds_2}x (If Match 1 slip survives)")
    print(f" > MATCH 3: Bet Php {h3_b:,.2f} @ {odds_3}x (If Match 2 slip survives)")

else:
    print("MATHEMATICALLY IMPOSSIBLE")
    shortfall = min_fullslip_profit - max_possible_fullslip_profit
    print(f"You are asking for Php {shortfall:,.2f} more than the odds mathematically allow.\n")

    print("COMPROMISE OPTIONS")
    print(f"Option A (Prioritize Hedge Floor):")
    print(f"If you strictly secure a hedge profit of Php {min_hedge_profit:,.2f},")
    print(f"the MAXIMUM possible slip profit is Php {max_possible_fullslip_profit:,.2f}.")
    print(f"Total Hedge Cost: Php {total_hedge_budget:,.2f}\n")

    print(f"Option B (Prioritize Slip Floor):")
    print(f"If you strictly secure a slip profit of Php {min_fullslip_profit:,.2f},")
    print(f"the MAXIMUM possible hedge profit is Php {max_possible_hedge_profit:,.2f}.")
    print(f"Total Hedge Cost: Php {max_budget_allowed:,.2f}")
```

Listing 4: Target-Bound Split Optimization

Here is the tabulated output of Strategy C:

Scenario A: Maximize Slip Upside

By limiting the hedge safety net to a strict minimum profit of Php 3,000.00, the remaining asset value is released to maximize the primary parlay payout, pushing the slip profit to **Php 22,991.52** requiring a total hedge budget allocation of Php 6,868.92.

Step	Target	Hedge Amount	Odds	Conditional Trigger Condition	Net Profit
1	Match 1 (England)	Php 608.89	6.000x	England fails to win group stage	**Php 3,000.00**
2	Match 2 (Portugal)	Php 3,727.89	1.980x	England wins; Portugal fails to win match	**Php 3,000.00**
3	Match 3 (Austria)	Php 2,532.15	3.915x	Portugal wins; Austria fails to secure 2nd	**Php 3,000.00**
4	Full	—	—	Austria secures 2nd (Parlay successfully clears)	**Php 22,991.52**

Table 3: Target-Bound Split Optimization (Scenario A)

Scenario B: Maximize Hedge Safety Net

By fixing the primary parlay profit to a strict target floor of Php 20,000.00, the algorithm redirects all surplus capital to maximize the defensive safety cushion, increasing the equalized fallback hedge profit to **Php 4,325.90** with a total hedge cost of Php 9,860.44.

Step	Target	Hedge Amount	Odds	Conditional Trigger Condition	Net Profit
1	Match 1 (England)	Php 874.07	6.000x	England fails to win group stage	**Php 4,325.90**
2	Match 2 (Portugal)	Php 5,351.44	1.980x	England wins; Portugal fails to win match	**Php 4,325.90**
3	Match 3 (Austria)	Php 3,634.94	3.915x	Portugal wins; Austria fails to secure 2nd	**Php 4,325.90**
4	Full	—	—	Austria secures 2nd (Parlay successfully clears)	**Php 20,000.00**

Table 4: Target-Bound Split Optimization (Scenario B)

Action Plan Summary (Strategy C):

- **Scenario A Route:** Place an initial hedge of **Php 608.89** on Match 1. If it hits, collect your baseline profit and *STOP*. If it loses, conditionally allocate **Php 3,727.89** to Match 2, and then **Php 2,532.15** to Match 3 if the slip continues to survive.
- **Scenario B Route:** Place a heavier defensive hedge of **Php 874.07** on Match 1. If it hits, collect your maximized safety net and *STOP*. If it loses, conditionally allocate **Php 5,351.44** to Match 2, and then **Php 3,634.94** to Match 3 if the slip continues to survive.

4.4 Strategy D: Budget-Capped Hedging

```
# Your new target payout if the slip wins
target_fullslip_payout = 20000.00
# The maximum total capital you are willing to spend on hedges
max_hedge_budget = full_payout - target_fullslip_payout

# PARTIAL HEDGING ALGEBRA
# We calculate the required ratios between the bets to equalize the payout
# at whatever stage the slip fails.
ratio_1 = 1.0
```

```
ratio_2 = odds_1 / (odds_2 - 1)
ratio_3 = ratio_2 * (odds_2 / (odds_3 - 1))

total_ratio = ratio_1 + ratio_2 + ratio_3

# Distribute the 9,904.88 budget according to these mathematical ratios
hedge_1 = (ratio_1 / total_ratio) * max_hedge_budget
hedge_2 = (ratio_2 / total_ratio) * max_hedge_budget
hedge_3 = (ratio_3 / total_ratio) * max_hedge_budget

# CALCULATE NET PROFITS
profit_1 = (hedge_1 * odds_1) - hedge_1 - initial_stake
profit_2 = (hedge_2 * odds_2) - hedge_1 - hedge_2 - initial_stake
profit_3 = (hedge_3 * odds_3) - hedge_1 - hedge_2 - hedge_3 - initial_stake
profit_slip = target_fullslip_payout - initial_stake

print("PARTIAL SEQUENTIAL HEDGING")
print(f"Goal: Secure exactly Php {target_fullslip_payout:,.2f} payout if the slip
    wins,")
print(f"while maximizing the guaranteed profit if the slip fails.\n")

print("ACTION PLAN")
print(f"STEP 1: MATCH 1 (England)")
print(f"Action: Place a hedge of Php {hedge_1:,.2f} at {odds_1}x odds.")
print(f"If it hits: Profit = Php {profit_1:,.2f}")
print(f"If it loses: Proceed to Step 2.\n")

print(f"STEP 2: MATCH 2 (Portugal)")
print(f"Action: Place a hedge of Php {hedge_2:,.2f} at {odds_2}x odds.")
print(f"If it hits: Profit = Php {profit_2:,.2f}")
print(f"If it loses: Proceed to Step 3.\n")

print(f"STEP 3: MATCH 3 (Austria)")
print(f"Action: Place a hedge of Php {hedge_3:,.2f} at {odds_3}x odds.")
print(f"If it hits: Profit = Php {profit_3:,.2f}")
print(f"If it loses: Slip Wins. Payout = Php {parlay_payout - hedge_1 - hedge_2 -
    hedge_3:,.2f} (Profit = Php {profit_slip:,.2f})")
```

Listing 5: Budget-Capped Hedging

Here is the tabulated output of Strategy D:

Step	Target	Hedge Amount	Odds	Conditional Trigger Condition	Net Profit
1	Match 1 (England)	Php 878.01	6.000x	England fails to win group stage	**Php 4,345.60**
2	Match 2 (Portugal)	Php 5,375.55	1.980x	England wins; Portugal fails to win match	**Php 4,345.60**
3	Match 3 (Austria)	Php 3,651.32	3.915x	Portugal wins; Austria fails to secure 2nd	**Php 4,345.60**
4	Full	—	—	Austria secures 2nd (Optimized parlay payout target met)	**Php 19,955.56**

Table 5: Budget-Capped Hedging

Action Plan Summary (Strategy D):

- **Step 1 (England):** Place an initial hedge of **Php 878.01** at $6x$ odds. If it hits, collect your equalized profit of Php 4,345.60 and *STOP*. If it loses, the slip survives; proceed to Step 2.
- **Step 2 (Portugal):** Place a conditional hedge of **Php 5,375.55** at $1.98x$ odds. If it hits, collect your equalized profit of Php 4,345.60 and *STOP*. If it loses, the slip survives; proceed to Step 3.
- **Step 3 (Austria):** Place a final conditional hedge of **Php 3,651.32** at $3.915x$ odds. If it hits, collect your equalized profit of Php 4,345.60. If it loses, the primary slip successfully hits its targeted payout value, securing a net profit of **Php 19,955.56**.

4.5 Conclusion

This very simple project shows how to use simple logic and basic coding to completely take the luck out of risky, step-by-step situations. Because the remaining matches played out one after the other instead of all at once, I didn't have to risk all hedging money upfront.

The Python script gave me four clear choices on how to handle the wager depending on how safe I wanted to be:

- **Absolute Guaranteed Equal Profit:** Balanced the math so that no matter who won, lost, or drew, I walked away with the exact same profit.
- **Confidence-Weighted Manual Sizing:** Let me hedge less on teams that have higher implied probability of winning and more on teams that have lower.
- **Target-Bound Split Optimization:** Checked if my targets were possible and showed me how to get the highest payout while keeping a basic safety net.
- **Budget-Capped Hedging:** Lowered my main win payout so I could guarantee a solid hedge prize if the slip busted early.